고려대학교 약학대학
KOREA UNIVERSITY COLLEGE OF PHARMACY

KU-The Future
110 years and beyond

Microbiology



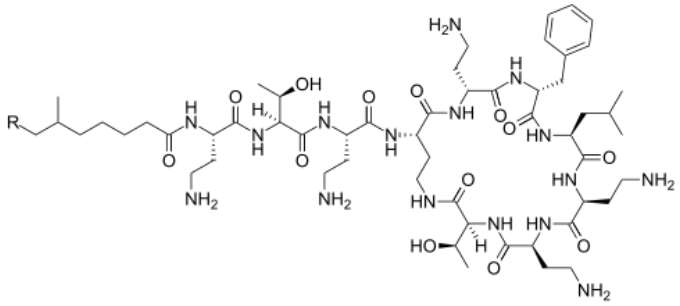
Fall semester, 2026

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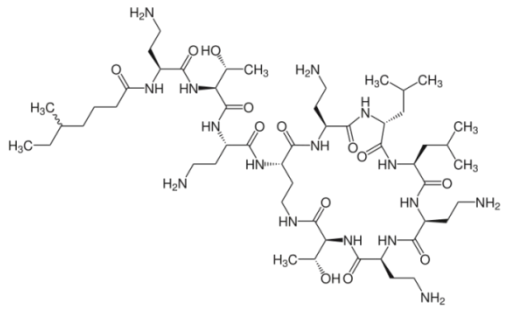
❖ Cell membrane disrupters

▶ Polymyxins

- cyclic **polypeptide antibiotics** with a long hydrophobic tail
- five polymyxins: A, B, C, D, and E (**colistin**)
  - B and E are the most common clinically
- produced from the soil bacterium *Bacillus polymyxa* or *Bacillus colistinus*
- usually used only as a last resort (last-line antibiotics)
  - intramuscularly or intravenously
  - be problematic because they have not been evaluated with the rigor of the modern drug approval process
  - strong adverse effects: nephrotoxicity and neurotoxicity



Polymyxin B



Polymyxin E

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❖ Cell membrane disrupters

▶ Polymyxins

· be usually applied to treat infections caused by G (–)
(e.g. *Pseudomonas* – especially, antibiotics-resistant)
– less effective against G (+)

· mechanisms of action:
– binding to lipopolysaccharide (LPS) in the outer membrane
: selectively toxic for G (–) bacteria
– causing both outer and inner membrane damage by the hydrophobic tail
– a detergent-like mode of action: increasing the membrane permeability

▶ Gramicidin: A, B and C

· a polypeptide antibiotic (mainly hydrophobic, but having some (+) charged AAs); produced from a soil bacterium (*Bacillus brevis*)

· spectrum of activity: G (+) and selective G (–) (e.g. Neisseria bacteria)

· strong toxicity (e.g. hemolysis); limited to topical application

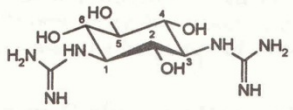
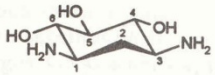
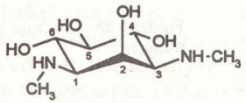
· mode of action: increasing the permeability of the bacterial cell membrane

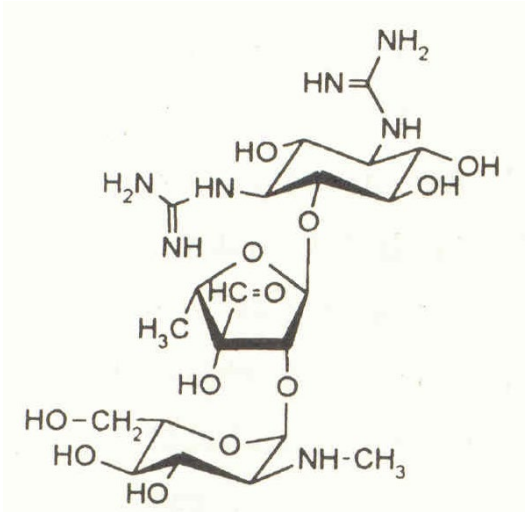
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❖ Protein synthesis inhibitors

▶ Aminoglycosides

- a molecule composed of amino-modified sugars
  : aminocyclitol + amino sugar (by glycosidic linkage)
- be derived from various species of the genera *Streptomyces* (-mycin; ex: streptomycin) and *Micromonospora* (-micin; ex: gentamicin)

Aminocyclitol		Main antibiotics
Name	Structure	
Streptidine		streptomycin
2-Deoxystreptamine		neomycin paromomycin ribostamycin kanamycin tobramycin gentamicin sisomicin amikacin dibekacin netilmicin
Actinamine		spectinomycin



Streptomycin

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❖ Protein synthesis inhibitors

▶ Aminoglycosides

· Classification:

1. Streptomyces-derived:
   - · natural: streptomycin, spectinomycin, neomycin, paromomycin, lividomycin A, ribostamycin, kanamycin, tobramycin, etc.
   - · semisynthetic: dibekacin, arbekacin, etc.
2. Micromonospora-derived:
   - · natural: gentamicin, sisomicin, etc.
   - · semisynthetic: isepamicin, netilmicin, etc.

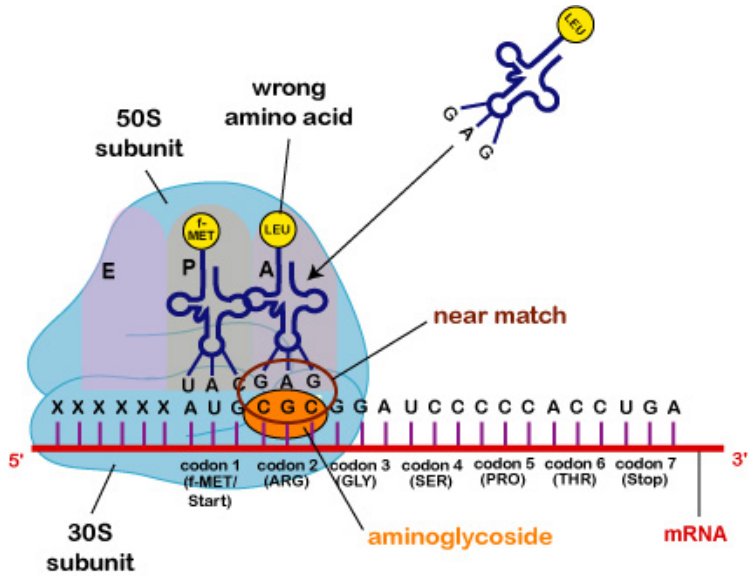
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❖ Protein synthesis inhibitors

▶ Aminoglycosides

· Mode of action

: the exact mechanism of action is not fully known.



– potential antibiotic mechanisms:

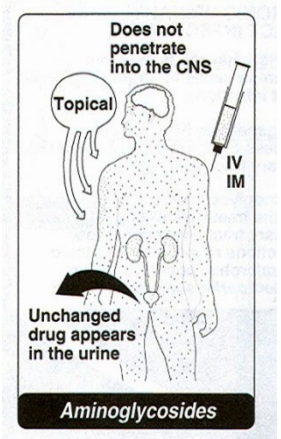
· irreversible binding to the 30S subunits of bacterial ribosomes
→ inhibition of polysome formation of ribosome
→ induction of misreading of genetic information (mRNA)
→ interfering with the proofreading process
→ inhibition of ribosomal translocation

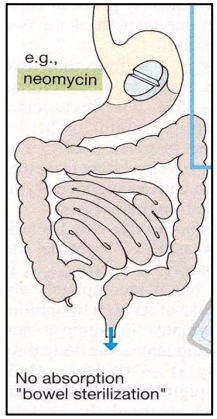
· disrupting the integrity of bacterial cell membrane

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❖ Protein synthesis inhibitors

▶ Aminoglycosides





- **Administration**:
  - be not absorbed from the gut
  - be administered intravenously and intramuscularly (not orally; except neomycin)
  - neomycin: very toxic; used only by oral or topical administration
- **Spectrum of activity**: broad-spectrum (G(−) > G(+))
  - be particularly active against most G (−) bacteria, including the enteric GNR (*Y. pestis*), the genus *Pseudomonas*, etc.
  - be also active against some G (+) bacteria, including MRSA, but not considered primary agents
  - combination with a cell wall-active agents: able to produce synergistic bactericidal effects (against *Streptococci* and *Enterococci*)
  - inherently lacking activity against anaerobic bacteria

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❖ Protein synthesis inhibitors

▶ Aminoglycosides

- Adverse effects:
  - nephrotoxicity (신독성) and ototoxicity (이독성):
    - neomycin: the most severely nephrotoxic and ototoxic
    - streptomycin: weakly nephrotoxic or ototoxic
    - spectinomycin: no serious adverse effects have not been reported / bacteriostatic (not bactericidal)
  - to reduce the risk
    - correct dosing and extended-interval dosing
    - avoidance of co-administration of other nephrotoxins (e.g. cyclosporin, etc.)
    - avoidance of co-administration of other nephrotoxic or ototoxic antibiotics (e.g. vancomycin, bacitracin, etc.)
  - originally, developed to use twice or three times a day, but recommended once a day

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❖ Protein synthesis inhibitors

▶ Aminoglycosides

· **mechanisms for inducing resistance:**

1. reduced uptake or decreased cell permeability
   · *P. aeruginosa* and other G (–) exhibit resistance due to a transport defect or membrane impermeabilization (likely chromosomally mediated).

2. enzymatic inactivation or modification
   : modification of the amino or hydroxyl groups of amino-glycosides by aminoglycoside-modifying enzymes

   · acetyltransferase: acetylation of amino group
   · adenyltransferase: adenylation of hydroxyl group
   · phosphotransferase: phosphorylation of hydroxyl group

3. altered ribosome binding sites
   · mutations at the site of aminoglycoside attachment on the ribosome (not common; ex: streptomycin)

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❖ Protein synthesis inhibitors

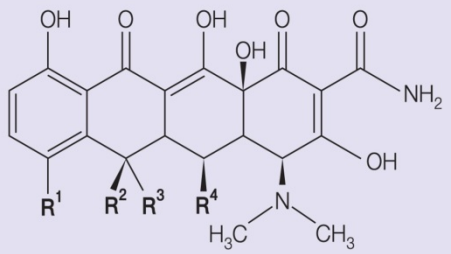
▶ Tetracyclines

· name: originated from their chemical structure (four-ring)

· be obtained from the genus *Streptomyces*
  – the first tetracycline: chlortetracycline (*S. aureofaciens*)

· representative broad-spectrum antibiotics: G (+)/(−); including atypicals, richettsia, etc.

· classification:
  – short-acting: chlortetracycline, oxytetracycline, tetracycline, etc.
  – intermediate-acting: demeclo-cylcline, methacycline, etc.
  – long-acting: doxycycline, minocycline, etc.
  → semi-synthetic: methacycline, doxycycline, minocycline, etc.
  → tigecycline: effective to tetra-cycline-resistant bacteria



	R^1	R^2	R^3	R^4
클로르테트라사이클린(chlortetracycline)	Cl	CH_3	OH	H
옥시테트라사이클린(oxytetracycline)	H	CH_3	OH	OH
테트라사이클린(tetracycline)	H	CH_3	OH	H
데메클로사이클린(demeclocycline)	Cl	H	OH	H

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❖ Protein synthesis inhibitors

▶ Tetracyclines

· mode of action:

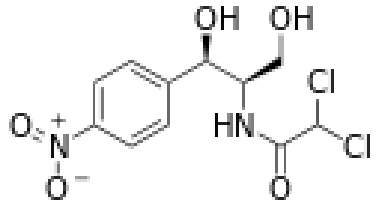
- irreversibly binding to the 30S subunits of bacterial ribosomes
- inhibition of binding of aminoacyl-tRNA to A site

· adverse effects:

- discoloration of developing teeth (yellowish teeth)
  : exception – oxytetracycline
- indigestion, nausea, and diarrhea: common in oral administration
- photosensitivity
- pustules on the skin
- kidney damage: leading to metabolic acidosis (대사성 산성증) and the excretion of proteins and glucose
- liver damage: fatal

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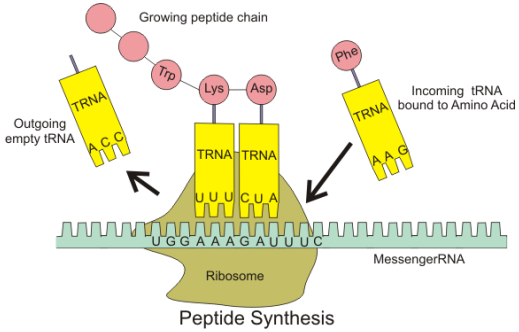
❖ Protein synthesis inhibitors



▶ Chloramphenicol

- one of the broad-spectrum antibiotics, alongside the tetracyclines
  : be active against G (+; including MRSA) and (−), etc.
- be obtained from the genus *Streptomyces* (*S. venezuelae*)
- be both cheap and easy to manufacture
  - introduced into clinical practice in 1949
  - (due to resistance and safety concerns) no longer a first-line agent for any infection in developed countries
  - be still widely used in low-income countries
- <u>adverse effects</u>:
  - bone marrow toxicity: suppression of bone marrow
  - aplastic anemia: rare, unpredictable, unrelated to dose, and in general fatal
  - leukemia

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❖ Protein synthesis inhibitors

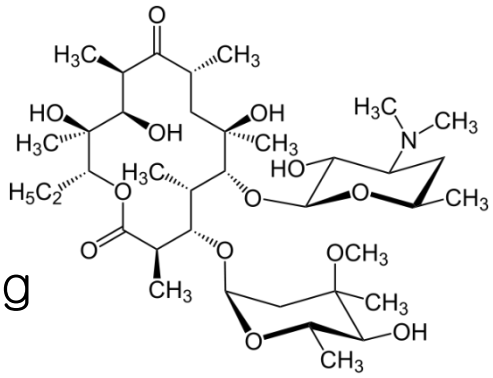
▶ Chloramphenicol

· Mechanism of action
: irreversibly binding to the P site on the 50S subunit of ribosome (especially, 23S ribosomal RNA)
- inhibiting the peptidyl transferase activity of ribosome (actually, by rRNA)
  → forming peptide bonds between adjacent amino acids
- premature dissociation of the peptidyl-tRNA from the ribosome

· Mechanisms of resistance:
- mutation of the 50S ribosomal subunit
- chloramphenicol acetyltransferase: covalently attaching an acetyl group from acetyl-CoA to chloramphenicol
- reduced membrane permeability: produced easily by serial passage of bacteria

❖ Protein synthesis inhibitors

▶ Macrolides (antibacterial)



Erythromycin

· be composed of a large macrocyclic lactone ring + one or more deoxy sugars
→ named macrolide (large-ring compound)

· lactone ring: 14-, 15-, or 16-membered

- Erythromycin:
  · natural product by *Streptomyces erythreus*
  · 14-membered; commonly used
- Semisynthetic products
  · 14-membered: Clarithromycin (Biaxin), Dirithromycin, etc.
  · 15-membered: Azithromycin (Zithromax)
  · 16-membered: Spiramycin, Josamycin, etc.
  · **Ketolides**: derivative of erythromycin by replacing the cladinose (visible at the bottom) with a keto group (R-CO-R') / telithromycin (only a ketolide on the market), etc.

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❖ Protein synthesis inhibitors

▶ Macrolides (antibacterial)

· Spectrum of activity:
- having broad coverage of respiratory pathogens (e.g. *S. pneumoniae*, *Haemophilus influenzae*, etc.)
  → one of the antibiotics used most frequently in the outpatient setting
- be slightly wider than that of penicillin
  → a common substitute for patients with a penicillin allergy
- coverage being not particularly deep: bacteriostatic

· Adverse effects:
- gastrointestinal: nausea, vomiting, diarrhea, etc.
- hepatic: rare but serious; hepatic failure leading to death or the need for transplantation
- cardiac: prolongation of the QT intervals; use with caution in patients with preexisting heart conditions
- increasing resistance: especially in *S. pneumoniae*

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❖ Protein synthesis inhibitors

▶ Macrolides (antibacterial)

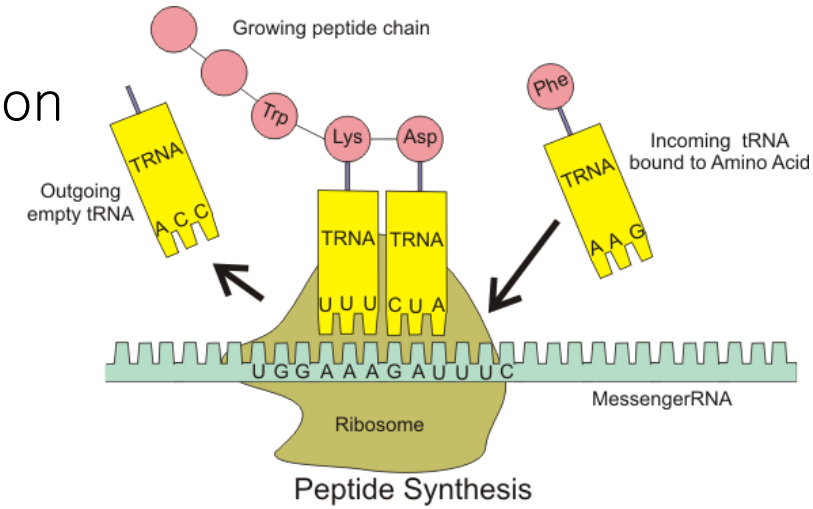
· Mechanism of action (similar to chloramphenicol):

– irreversibly binding to the P site on the 50S subunit of ribosome (especially, 23S ribosomal RNA)

· inhibting the peptidyl transferase activity of ribosome (actually, by rRNA)
→ forming peptide bonds between adjacent amino acids

· premature dissociation of the peptidyl-tRNA from the ribosome

– by inhibiting ribosomal translocation



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❖ Protein synthesis inhibitors

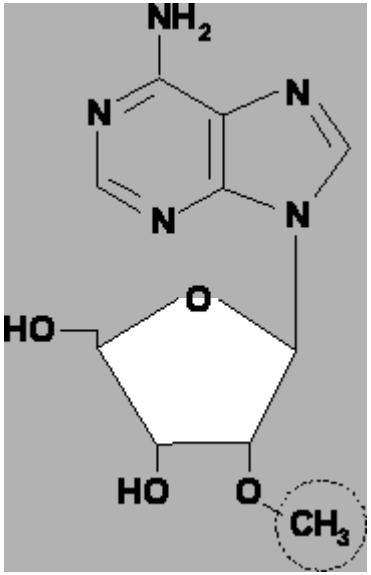
▶ Macrolides (antibacterial)

· Mechanism of resistance:

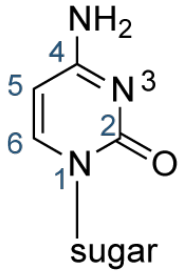
- primarily occurs by post-transcriptional methylation of the 23S ribosomal RNA
- rarely occurs by the production of drug-inactivating enzymes (e.g. esterases or kinases)
  · esterase: ester (R-COO-R') + H_2O → alcohol + acid
  · kinase: enzyme transferring phosphate groups
- rarely occurs by the production of active ATP-dependent efflux proteins

- Protein synthesis inhibitors
- Macrolides (antibacterial)

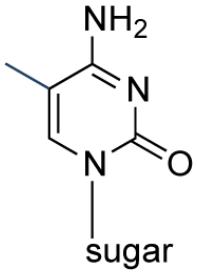
rRNA modifications



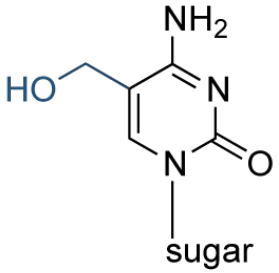
ribose methylation



cytosine
(C)

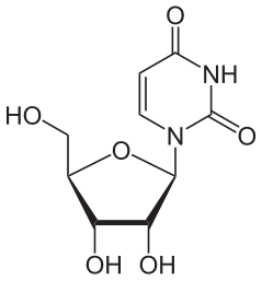


5-methylcytosine
(mC)

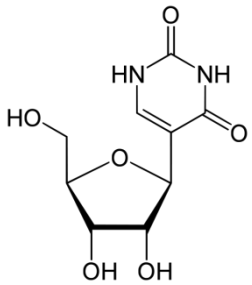


5-hydroxymethylcytosine
(hmC)

Cytosine methylation



Uridine



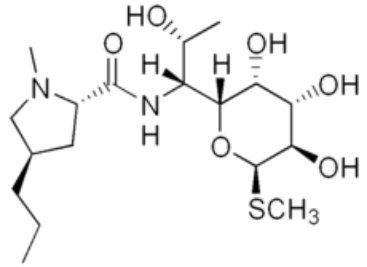
Pseudouridine

Pseudouridylation

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❖ Protein synthesis inhibitors

▶ Lincosamides



Lincomycin

· Bacteriostatic or bacteriocidal antibiotics

· Two representative members: lincomycin and clindamycin
  – lincomycin: natural product from *Streptomyces lincolnensis* (first isolated in Lincoln, Nebraska, USA)
  – clindamycin: semisynthetic product of lincomycine
  – clindamycin is more completely absorbed and less toxic (having improved antibacterial activity)

· Spectrum of activity:
  – active against many G (+) and some G (–) bacteria
  – active against staphylococci (including MRSA), streptococci, etc.
  – active against parasitic protozoa: e.g. *Toxoplasma*, *Plasmodium* species (malaria)

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❖ Protein synthesis inhibitors

▶ Lincosamides

· Adverse effects:
  – gastrointestinal: e.g. diarrhea
  – dermatologic: e.g. skin rash

· Mechanism of action (be similar to macrolides)
  – irreversibly binding to the P site on the 50S subunit of ribosome (especially, 23S ribosomal RNA)
    · inhibiting the peptidyl transferase activity of rRNA
    · premature dissociation of the peptidyl-tRNA from the ribosome

· Mechanism of resistance (be similar to macrolides)
  – altering the drug's binding site
  – enzymatic inactivation of agents